

15.3.1: The Terminology of Hypothesis Testing:

Information can be inferred about the population through statistics obtained by samples. A hypothesis is used to test the validity of these claims.

Format for a Hypothesis Test:

1) Define the variables being investigated
2) Formulate the Null ($\rho = 0$) & Alternate Hypotheses. Say what ρ is, a population PMCC / an association
3) Declare tailedness of test & significance level, with population params. Sig. Lvl. is usually 5%
4) Determine the critical value & region, and find a sample value for ρ (test statistic)
5) Determine whether the test statistic is in any critical region.
6) Confirm validity of H_0.
7) Interpret findings, but only suggest!

Notes:

Significance level is % chance of incorrectly rejecting H_0.
Test statistic must be obtained from the inferred sample
Critical Values are boundaries of critical regions.
P-Value is the probability of the variable being more extreme than the test statistic, towards the nearest definite
Tailness can be determined by a question's wording
Sample must be obtained randomly, else everything is invalid.

Result Inferences:

Accept H_0: The test statistic is not in the critical region
Reject H_0: The test statistic is in the critical region

Significance Level:

The advantage of a lower significance is that there is a smaller chance of incorrectly rejecting H_0. The disadvantage is hence a higher chance of incorrectly accepting H_0.

15.3.2: Hypothesis Testing about the Population PMCC

H_0 (Null Hypothesis) suggests: $\rho = 0$
H_1 (Alternate Hypothesis) may suggest: $\rho > 0$, $\rho < 0$, $\rho \neq 0$
(One-tailed, One-tailed, two-tailed)

The test statistic is the sample PMCC value which is given in the question.
The Critical Values are the minimum level of correlation required to be concluded a correlation is present.
The critical values are dependent on how many tails there are

15.3.2: Hypothesis Testing about The Population PMCC, CONTINUED:

Notes:
When this hypothesis test is performed, it is assumed that the underlying population must have bivariate Normal Distribution
An advantage over SRCC is that actual data is used & is more likely to spot underlying correlation

Critical Regions



One tailed Positive

One tailed Negative

Two-tailed

PMCC Format:
H_0: $\rho = 0$
H_1: $\rho \neq 0$, $\rho > 0$, $\rho < 0$
} where ρ is the population PMCC value between _____ & _____.
$r = x$ (PMCC value)
One / Two-tailed, n = [Sample Size], Significance level = y% $\Rightarrow CV^+$ = _____ } Sometimes only 1 needed
Critical Region: $r \geq CV^+$ ∪ $r \leq CV^-$ (2T ONLY!) CV^- = _____ }
If r is in the critical region, reject H_0. Else, do not.
Therefore, there is / is not sufficient evidence to suggest (Positive / Negative) Correlation between _____ & _____.

15.3.3: Hypothesis Testing about the SRCC:
SRCC is NOT a population parameter as it uses ranks of original data, hence there is no specific notation as we are testing the association of the data.

H_0 (Null Hypothesis) suggests: there is no association between _____ & _____
H_1 (Alternate Hypothesis) suggests: there is a (positive / negative) association between _____ & _____

The test statistic is the Sample SRCC
The critical values are the level of association required to be convinced an association is present

Does not require underlying bivariate normal distribution in the populations.
Same method as PMCC, just use correct terminology.

Hypothesis Testing with the Binomial Distribution

18.1: Terminology of Hypothesis Testing

A hypothesis test is used to test validity of inferences about the population from statistics
Follow the normal hypothesis test method, remember to define variables.
Binomial testing finds the probability of the event, or more extreme, occuring under the initial presumed probability. If this is less than the sig. lvl., H_0 is rejected. Hence $X \sim B(n, p_0)$, where X is the no. of successes

Null: $H_0: p = p_0$

Alternate: $H_1: p > p_0$ $H_1: p \neq p_0$
$H_1: p < p_0$

Critical Regions



One tailed
$H_1: p > p_0$

One tailed
$H_1: p < p_0$

Two-tailed
$H_1: p \neq p_0$

If the test statistic lies in a critical region, reject H_0. Else, do not.
Samples must be fully random, and hypotheses cannot be changed after sample data is collected.

Using p-values:

$H_0: p = p_0$
$H_1: p \neq p_0$ (or other) } where p is the prob of...

Two-tailed, n = Sample size, $\alpha = \frac{\text{Sig. lvl.}}{2}$ (halved for 2 tails!)

$X \sim B(n, p_0)$. Then find $P(X \geq x)$ or $P(X \leq x)$. If these are below sig lvl., reject H_0.

Using Critical Regions

H_0: $p = p_0$
H_1: $p \neq p_0$ (or other) } where p is the prob of...

Two-tailed, n = Sample Size, $\alpha = \frac{\text{Sig. Lvl.}}{2}$ (halved for 2 tails!)

CV^+: $P(X \geq x_1) > \frac{\alpha}{2}$, $P(X \geq x_2) < \frac{\alpha}{2}$ } $CV^+ = x_2$ Using $\sum_{n=}^{n_2} P(X = x)$

CV^-: $P(X \leq x_3) > \frac{\alpha}{2}$, $P(X \leq x_4) < \frac{\alpha}{2}$ } $CV^- = x_4$

$\Rightarrow$ CR: Reject if: $x \leq x_4$, $x \geq x_2$

Can use inverse binomial on calculator, just check using Bcd afterwards!
If upper bound, find $100 - \alpha$.